

Network Traffic Analysis Using Wireshark

Investigating Suspicious Activity



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Investigation and Findings from Wireshark Packet Analysis

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July 23, 2026





Introduction

OBJECTIVE

- Analyze a captured network traffic file using Wireshark to identify and investigate potential security threats.

IMPORTANCE

- Network traffic analysis is essential for detecting abnormal activity, identifying security incidents, and supporting effective risk mitigation.

SCOPE

- Review network traffic to identify indicators of compromise (IoCs), assess potential false positives, and document key findings from the investigation.



Methodology

Step-by-Step Analysis Process

1. PCAP File Setup

- Loaded packet capture into Wireshark.
- Applied filters to isolate relevant traffic.

2. Initial Observations

- Identified key protocols:
 - DNS, TLS, QUIC, SMB, HTTP
- Reviewed overall traffic patterns.

3. Deep Packet Inspection

- Investigated notable traffic flows.
- Examined encrypted communications and protocol behavior.

4. Verification of Findings

- Evaluated potential false positives.
- Validated indicators using VirusTotal and IP analysis.

Methodology

PCAP File Setup

Actions Performed

- Loaded the Task #1003: Wireshark Packet Analysis packet capture file into Wireshark.
- Applied display filters to isolate relevant traffic.
- Prepared the dataset for analysis.

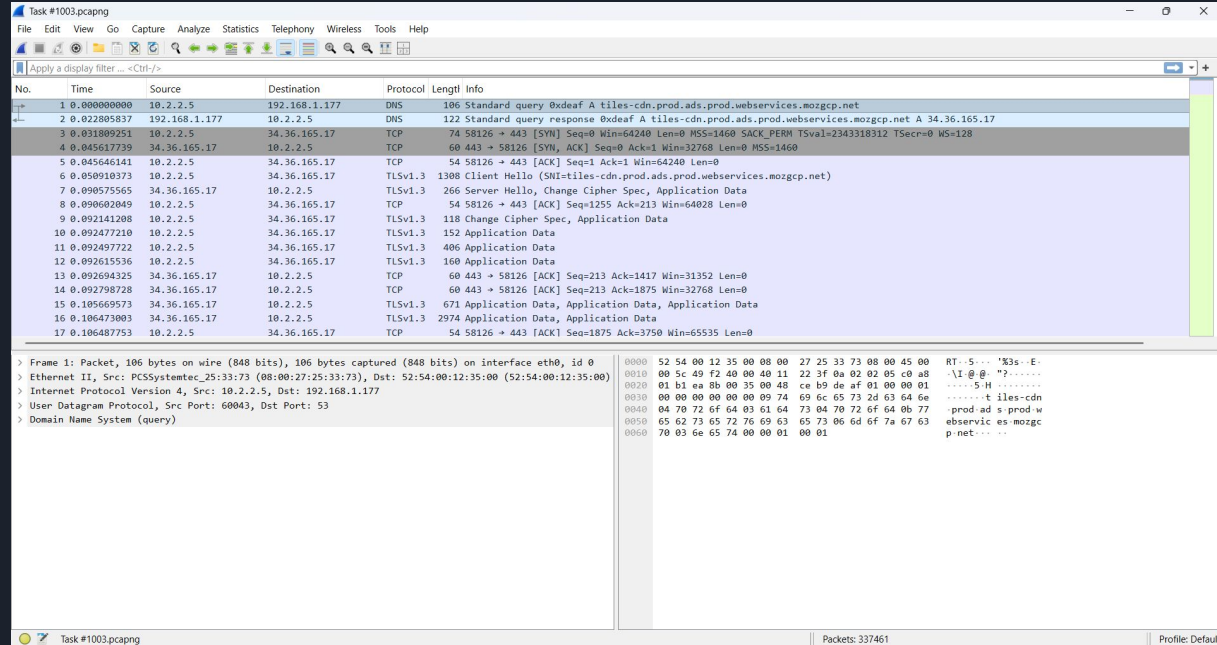
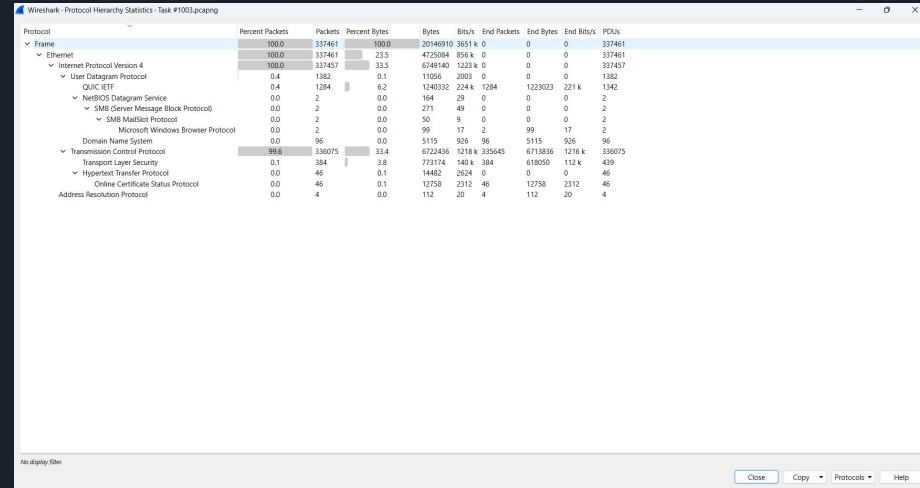
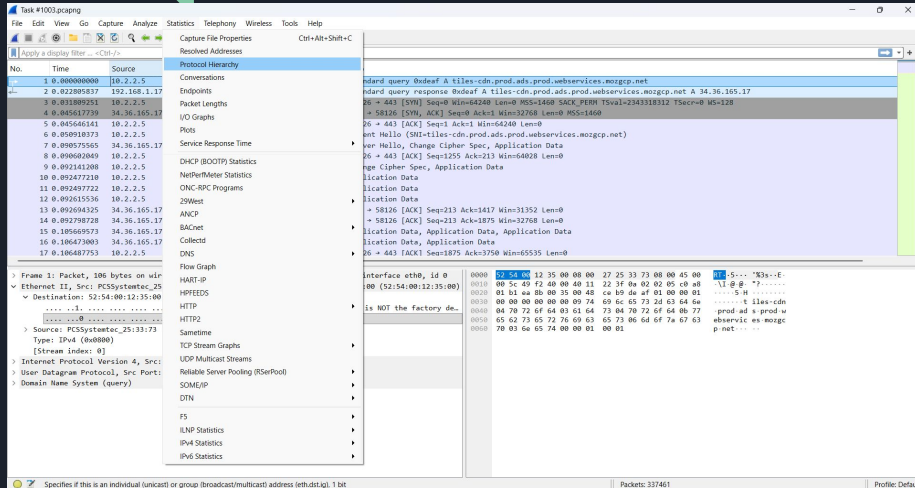


Figure 1 demonstrates the initial setup of the Wireshark analysis environment, showing the Task #1003 (Wireshark Packet Analysis) file successfully loaded and the captured network traffic ready for investigation.

Methodology

Initial Observations



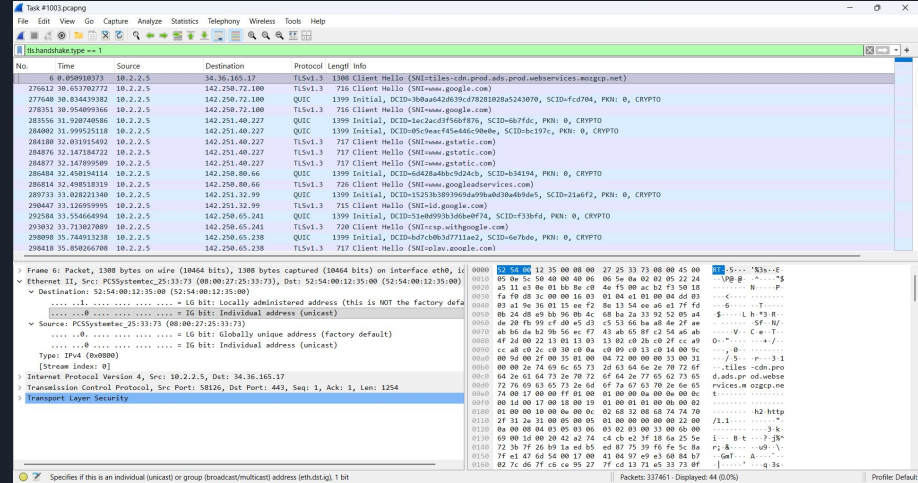
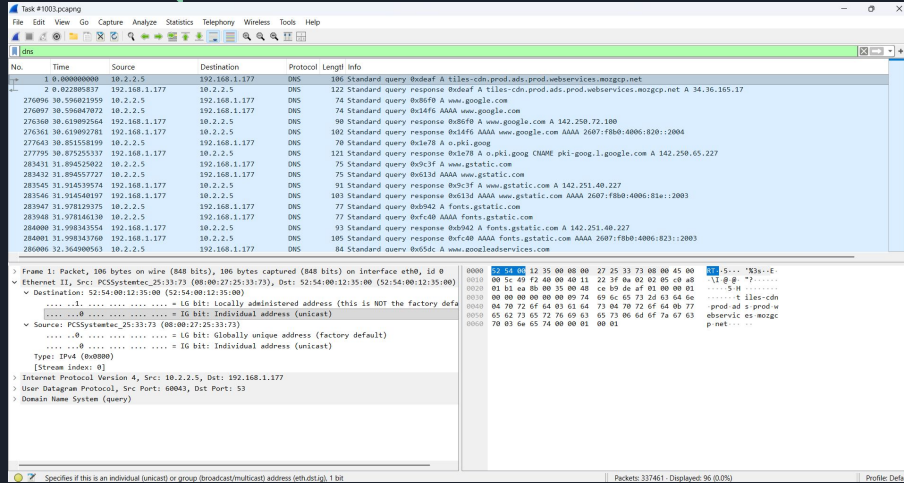
Protocol Analysis

- Reviewed protocol distribution using Wireshark's Protocol Hierarchy Statistics.
- Identified key traffic types, including DNS, TLS, QUIC, SMB, and HTTP.
- Established a baseline of network activity to guide deeper analysis.

Figures 2 and 3 demonstrate the use of Wireshark's Protocol Hierarchy analysis, showing the process of accessing the feature and reviewing network distribution statistics.

Methodology

Deep Packet Inspection



Detailed Analysis (DNS, TLS Analysis)

- Analyzed DNS queries to identify domain activity and potential suspicious lookups.
- Investigated TLS handshakes to verify encrypted communications.
- Reviewed packet details to identify potential indicators of compromise (IoCs).

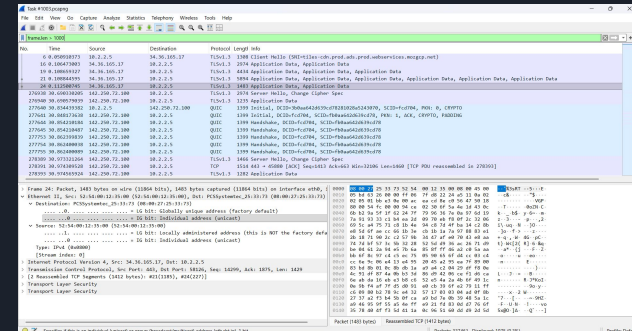
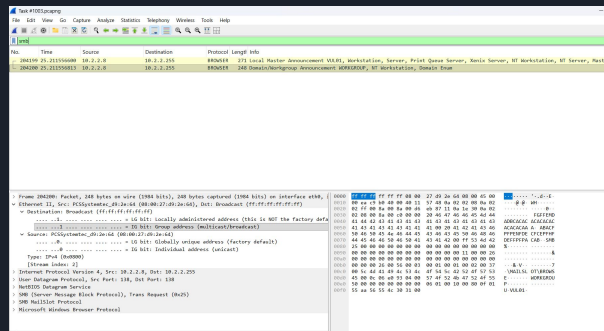
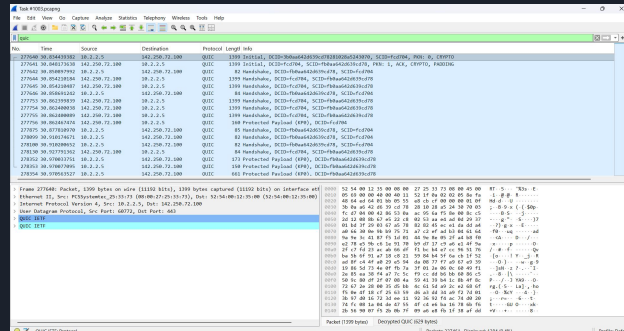
Figures 4 and 6 demonstrate the deep packet inspection process in Wireshark, showing the analysis of DNS queries and TLS handshake communications to identify network activity patterns and evaluate potential security concerns.

Methodology

Deep Packet Inspection

Detailed Analysis (QUIC, SMB, Traffic Analysis)

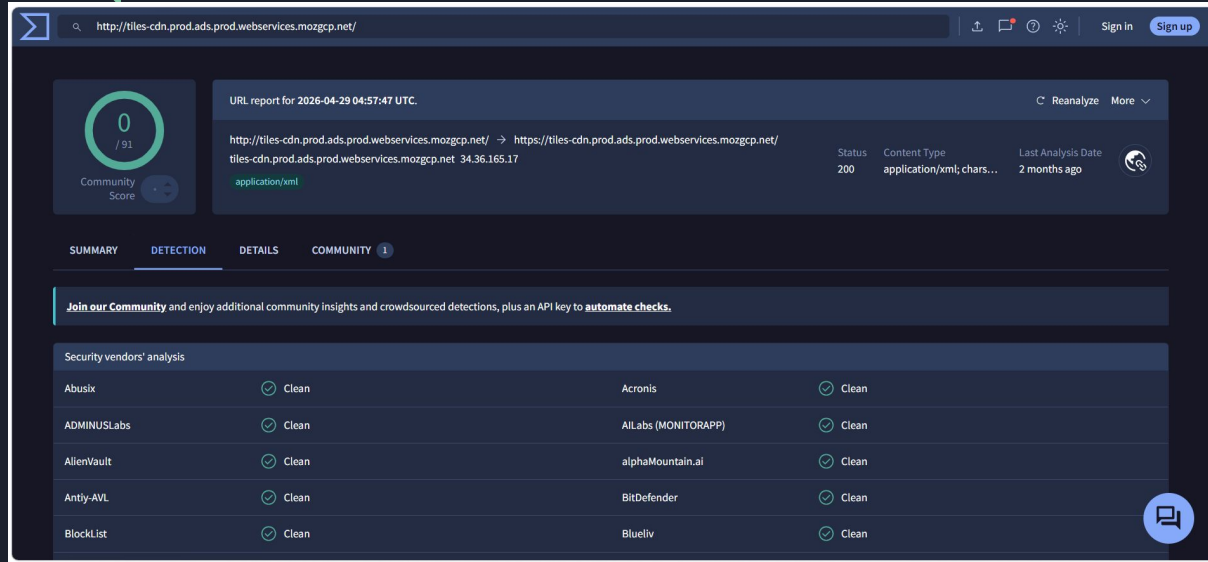
- Examined QUIC protocol behavior and encrypted UDP communications.
- Analyzed SMB file-sharing activity for potential anomalies.
- Reviewed packet sizes to identify unusual data transfers or possible exfiltration attempts.
- Evaluated traffic patterns to determine whether activity was legitimate or suspicious.



Figures 7–9 demonstrate the continued deep packet inspection process in Wireshark, showing the analysis of QUIC traffic, SMB communications, and packet size patterns to identify anomalies and evaluate potential security risks.

Methodology

Verification of Findings



Evaluated suspicious activity to eliminate false positives.

Correlated network indicators with external threat intelligence sources.

Validated suspicious domains using VirusTotal and reputation checks.

Confirmed findings before reaching final conclusions.

Figure 5 demonstrates the verification process using VirusTotal, showing the analysis of a suspicious domain to confirm its reputation, eliminate false positives, and validate the findings before reaching a final conclusion.



Methodology

Key Findings

DNS Analysis

- Normal DNS resolution patterns observed.
- No malicious domains or DNS tunneling activity detected.
- Suspicious domain verified as legitimate through VirusTotal.

TLS/HTTPS Analysis

- TLS 1.3 encrypted communications were properly established.
- No evidence of MITM attacks, invalid certificates, or suspicious endpoints.

QUIC Protocol Analysis

- Encrypted UDP traffic initially appeared suspicious.
- Traffic was confirmed to be associated with legitimate Google services.

SMB Traffic Analysis

- SMB traffic reflected normal Windows network discovery activity.
- No evidence of unauthorized access, lateral movement, or brute-force attempts.



Conclusion & Thank You

Final Take Away



Task #1003: Wireshark Packet analysis demonstrated the importance of a structured approach to network traffic investigation. Through protocol analysis, deep packet inspection, and threat intelligence validation, suspicious activity was investigated and confirmed to be legitimate. This process highlights the importance of evidence-based decision-making, accurate threat identification, and reducing false positives in cybersecurity operations.

Thank You